Predicting HDB Monthly Rents with a Two-Stage Tabular BERT

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***Abstract*—Monthly rent estimates help tenants and owners compare similar homes in Singapore's HDB rental market. Prediction is difficult because test approvals mostly occur after the labeled period and identical observed inputs can have different rents. We combine address normalization, rent cleaning and geographic features with a tabular BERT trained by masked-field reconstruction and rent regression. On a future-month holdout cleaned before splitting, BERT achieves 468.34 SGD/month RMSE against retained original rents, compared with 468.11 for shallow CatBoost and 497.97 for a recent grouped mean. This window shows little RMSE benefit from the more complex model, while separate baseline ablations with training-only cleaning identify the time group as the most influential tested feature group.**

I. INTRODUCTION

Monthly-rent estimates are useful when tenants compare similar homes, when owners or agents set an asking rent, and when analysts study how housing access changes over time. We formulate rent estimation as supervised regression in which most test approvals occur after the labeled training period. Evaluation must reflect this temporal gap to assess whether estimates remain useful for future rentals.

Three properties make this task harder than a random row split suggests. First, 49,620 of the 50,000 test rows occur after the last labeled month, with a maximum gap of 16 months; the problem includes temporal extrapolation. Second, the observed fields are coarse. In the raw training file, 17,900 groups share every recorded predictor field but have multiple rents, covering 39,540 rows. Floor level, condition and other unit-level attributes are unavailable, so identical inputs cannot receive different predictions. Third, address strings and auxiliary tables use heterogeneous representations: formatting variants affect joins, building facts contain missing values, and facility snapshots do not establish their historical availability. These properties motivate the preprocessing and temporal evaluation used below.

We address these issues with an auditable pipeline. It normalizes the core fields, applies a comparable-based rent cleaning rule while retaining the original label, joins each row to an HDB block and time-filtered MRT features, and encodes mixed fields as a fixed sequence. A randomly initialized BERT encoder then undergoes masked-field reconstruction without rent labels and is fine-tuned with a regression head. The reported selection run uses whole-month validation at the end of the labeled period; a full-data refit is trained only after the epoch counts have been selected. The model applies existing transformer ideas to rental records.

These observations lead to three research questions:

- **RQ1:** Under a future-month holdout, how much accuracy is gained when moving from a global mean and a recent grouped mean to Ridge, a shallow tree ensemble and the two-stage BERT?
- **RQ2:** Which feature groups matter most under a strict temporal protocol: location IDs, facility features or explicit time features?
- **RQ3:** Where do errors concentrate across months and building coverage, and what do these patterns show about temporal extrapolation and the limits of the available fields?

Sections VI and VII answer these questions with separate historical and strict protocols. We connect EDA findings to reproducible normalization, target cleaning, geographic enrichment and numeric encoding; document the two-stage field-token model and controlled baseline comparisons; and analyze temporal, building-level and within-input error.

II. TASK AND DATASET

The temporal and data-quality patterns in Section I determine how the prediction task is defined and which sources can be used without implying information that is unavailable at prediction time.

A. Prediction Task and Metric

Each row records an approved HDB rental application. Let x_i denote the property, location, facility and approval-month fields for row i, and let y_i be `MONTHLY_RENT` in Singapore dollars per month. The task is supervised regression, $\hat{y}_i = f(x_i)$, with the root mean squared error in Eq. 1:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2}, \tag{1}$$

computed on the original SGD/month scale with one equal weight per evaluated row. We also report mean absolute error (MAE) and mean error to make systematic under- or over-prediction visible. The local validation numbers below are not scores on the unlabeled competition test set.

TABLE I

CORE AND AUXILIARY SOURCES USED BY THE PIPELINE. COUNTS ARE ROWS IN THE LOCAL FILES; DATES ARE APPROVAL OR SOURCE COVERAGE WHERE AVAILABLE.

Source	Rows	Role and temporal scope
Train CSV	150,000	Labeled approvals, 2021-01–2025-03; 11 columns including rent
Test CSV	50,000	Prediction inputs, 2025-03–2026-07; 10 columns, no rent
HDB blocks	9,464	Address key, coordinates, completion/floor facts and planning areas
MRT stations	243	Coordinates and status; counts filtered by the opening calendar
Schools	337	Coordinates and school level for static distance/count features
Shopping malls	113	Coordinates for static distance/count features

B. Core and Auxiliary Data

Table I summarizes the core and auxiliary files. The core files contain 150,000 training rows and 50,000 test rows. Training approvals span 2021-01–2025-03 (51 observed months), whereas test approvals span 2025-03–2026-07 (17 observed months). Thus 49,620 test rows occur after the training maximum month and the maximum month index gap is 16; 380 test rows share the boundary month 2025-03 with training. The test file has no rent target. These counts describe records, not unique physical flats: the files contain repeated addresses and repeated feature combinations for different applications.

The core predictors include approval month, town, block, street, flat type, flat model, floor area and lease commencement year. `FURNISHED` is `yes` and `FEE` is zero in both splits, so the implementation audits them but excludes these constants from the model input. The HDB block table is joined on the normalized `(TOWN, BLOCK, STREET)` key. This key matches every train and test row to exactly one block record, with no one-to-many conflicts. The resulting block coordinates support Haversine distances and the facility counts described in Section IV.

The repository also contains Certificate of Entitlement and stock-price tables. They are not used in the reported model: their publication timing, historical snapshots and missing-value semantics are not sufficient for a clean fixed-origin feature protocol in the current implementation. Schools and malls are static snapshots whose temporal limitations are discussed in Section VII. The local files extend to July 2026, one month beyond the course description; we use the files as supplied.

III. EXPLORATORY DATA ANALYSIS

The task definition in Section II exposes two risks that must be measured before modeling: the target changes over calendar time, and the supplied fields do not identify every rental unit. EDA guides the preprocessing and evaluation decisions that address these risks.

A. Rent Variation and Temporal Change

Across the 150,000 raw training rows, mean rent is SGD 2,719.04/month, the median is SGD 2,700, and the interquartile range is SGD 2,100–3,200. The first and ninety-ninth percentiles are SGD 1,300 and SGD 4,600, while the full range is SGD 300–7,600. The monthly mean rises from SGD 2,048.90 in 2021-01 $(n = 3,129)$ to SGD 3,162.71 in 2025-03 $(n = 3,032)$. These sample summaries combine temporal changes, sample composition and unobserved unit attributes; they do not isolate a causal market trend.

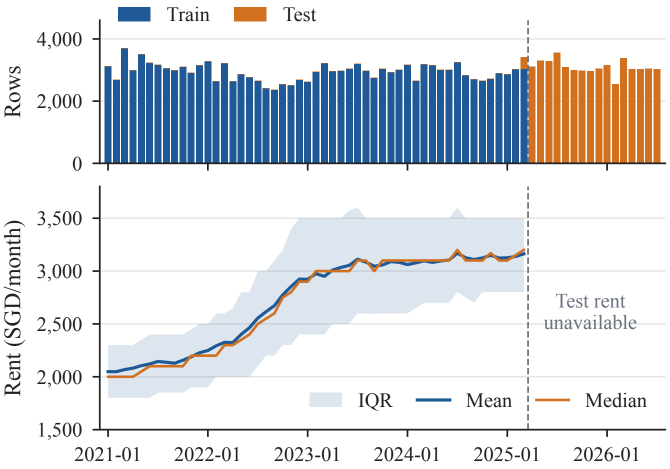

Fig. 1. Monthly train/test coverage (upper panel) and training rent summaries (lower panel). Rent is SGD/month; test rents are unavailable.

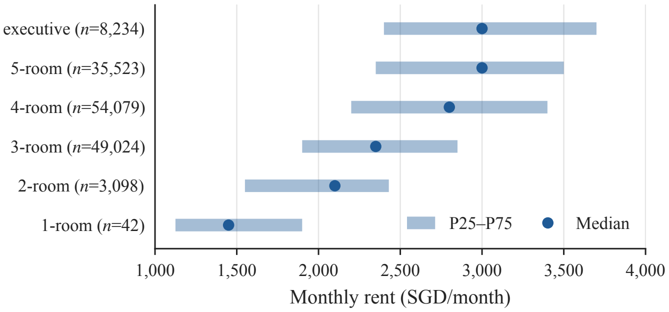

Fig. 2. Training rent distributions by normalized flat type, January 2021–March 2025. Horizontal bands show the observed interquartile range in SGD/month; sample sizes are unequal and the 1-room group is sparse.

Figure 1 shows that the later labeled months are not exchangeable with the earliest months. Because the test period is mostly later than 2025-03, this finding motivates a whole-month time split and explicit month features in Sections V and VI.

Flat type and area separate broad rent levels. After deterministic normalization, the median rents are SGD 2,350 for 3-room flats, SGD 2,800 for 4-room flats, and SGD 3,000 for both 5-room and executive flats. The 1-room group has only 42 observations, so its marginal summary is unstable. Pearson correlations with rent are 0.3293 for floor area and -0.1781 for the lease-based age proxy. The associations and overlapping interquartile ranges motivate preserving structural fields and allowing nonlinear effects without monotonic constraints.

B. Data Quality and Feature Implications

The raw strings contain formatting variants that affect both statistics and joins. Unicode NFKC normalization, trimming, lowercasing and whitespace compression reduce training `STREET` values from 1,146 to 576, affecting 30,781 rows; for example, *Ang Mo Kio Avenue 3* (424 rows) and *ang mo kio avenue 3* (1,590 rows) become one street. Mapping `N room` and `N-room` to the canonical `N-room` spelling reduces `FLAT_TYPE` from 11 raw values to six, affecting 35,331 rows. The normalized `(TOWN, BLOCK, STREET)` key then matches 100% of both splits to a unique HDB block, whereas raw key matching covers only 79.48% of train rows and 79.04% of test rows. This directly motivates deterministic normalization before joining auxiliary attributes in Section IV.

The inputs do not uniquely determine rent. In the raw training data, 17,900 groups share all observed predictor fields but have multiple rent values, covering 39,540 rows. Without a unit identifier, floor level or condition, a model must give the same prediction to all rows in such a group. We retain the rows, remove only the explicitly defined comparable-based outliers, and replace remaining same-unit same-month labels by their group mean, retaining their sampling weight as described in Section IV.

The quality audit also finds constant `FURNISHED` and `FEE` fields, 11 HDB blocks with missing maximum-floor and completion-year facts, and one training row with a lease-based age of -1. We keep the raw row trace, convert documented sentinel values to missing derived facts, add a missingness flag, and retain the negative-age observation for audit. Finally, the test set contains 49,620 rows after the training maximum month and 102 rows in 50 blocks not seen in the full training history. These facts motivate month-aware validation and the unseen-building and long-horizon checks in Sections VI and VII.

IV. DATA PREPROCESSING AND FEATURE ENGINEERING

The EDA decisions are implemented as three auditable preprocessing stages. They are separate from the two training stages: preprocessing creates numeric tables, while training first reconstructs masked fields and then fits the rent head.

A. Normalization and Rent Cleaning

Stage 1 applies Unicode NFKC, trimming, lowercase conversion and whitespace collapse to text columns, canonicalizes flat-type spellings, and preserves a `ROW_ID` for every source row. It converts documented numeric and sentinel encodings without fuzzy address matching. For target cleaning, comparables share the same town, block, street, flat type, flat model, floor area and lease commencement year. Their approval months must lie within ±2 months of the row being checked, and at least three comparables are required. A rent is flagged when it is more than twice or less than half the comparable median and differs from that median by at least SGD 800. The most extreme flagged value is removed first and the median is recomputed before the next removal.

This procedure removes 315 of the 150,000 training rows and leaves 149,685. For each remaining same-unit same-month group, the target is replaced by the group mean rounded to two decimals; the original value is retained as `MONTHLY_RENT_ORIGINAL`. This is label aggregation, not row deduplication. Under squared error, the mean is the minimizer for a group that receives one common prediction, while retaining every row preserves the original sampling weight. The historical BERT pipeline applies this full-file cleaning before the time split. Section VI distinguishes its retained-row scores from the training-only cleaning used for strict baselines.

B. Building and Facility Features

Stage 2 joins each rental to the HDB block table with a many-to-one check on normalized `(TOWN, BLOCK, STREET)`. The join adds planning geography, latitude, longitude, maximum floor and completion year. For each row we compute Haversine distances using an Earth radius of 6,371,008.8 m, the nearest MRT distance, the mean distance to the nearest three stations, counts within 500 and 1,000 m, school distances and counts, and mall distances and counts. MRT counts are evaluated using stations active from their first complete operating month in the explicit calendar. School and mall features come from static snapshots, so their historical availability is not established by the join itself.

All normalized train and test keys match exactly one HDB record. Stage 3 converts missing `MAX_FLOOR` and `YEAR_COMPLETED` sentinels to derived missing values. Completion year is imputed as lease year minus one, and maximum floor is filled from the street, subzone, town and then global median hierarchy; `HDB_INFO_MISSING` records which rows required this operation. The imputation is a building-level fallback and must not be interpreted as an apartment's actual floor.

C. Encoding and Transformation

Table II summarizes the 30 scalar inputs. Eight categorical fields become integer IDs with zero reserved for unknown values. Nine structural numeric fields and ten distance/count fields are standardized; the latter are first transformed with $\log(1+x)$ to reduce skew. Month index, lease-based age and completion age are derived from the approval date, and month-of-year is represented by sine and cosine. The calendar pair is presented to the model as one field token, so 30 scalars become 29 field tokens.

The field names in Table II also make the fate of each core column explicit. `TOWN` is normalized, used in the HDB join, and encoded as `TOWN_ID`. `STREET` is normalized before the exact `(TOWN, BLOCK, STREET)` join, retained as `STREET_ID`, and combined with `TOWN` and `BLOCK` into `BUILDING_ID`; the raw string is not passed to the model. `BLOCK` enters through the join and building ID, with no separate block input. `SUBZONE` and `PLANNING_AREA` become integer IDs from the joined HDB record, while `REGION` is encoded in the same way. Canonicalized `FLAT_TYPE`

TABLE II
FEATURE GROUPS AFTER STAGE 3 ENCODING.

Group	Count	Encoding and EDA response
Categorical IDs	8	Integer vocabularies; zero is unknown. The inputs are `TOWN_ID`, `STREET_ID`, `BUILDING_ID`, `SUBZONE_ID`, `PLANNING_AREA_ID`, `REGION_ID`, `FLAT_MODEL_ID` and ordinal `FLAT_TYPE_ORD`.
Structural numeric	9	z-score: area, lease year, ages, month index, coordinates and building facts.
Facility numeric	10	$\log(1+x)$ then z-score for MRT, school and mall distances/counts.
Calendar	2	Sine/cosine of month-of-year; one calendar token. Represents month-of-year seasonality.
Missingness flag	1	`HDB_INFO_MISSING` for imputed building facts.

becomes the ordered `FLAT_TYPE_ORD`, and `FLAT_MODEL` becomes `FLAT_MODEL_ID`.

The raw `RENT_APPROVAL_DATE` string is converted to `MONTH_INDEX`, `MONTH_SIN`, `MONTH_COS`, `AGE_YEARS` and `COMPLETION_AGE`, then dropped. `FLOOR_AREA_SQM` and `LEASE_COMMENCE_DATE` remain standardized numeric inputs. `FURNISHED` and `FEE` are constant in both splits and are dropped after the constant check. The cleaned `MONTHLY_RENT` is the training target; `MONTHLY_RENT_ORIGINAL` is retained only to score the preserved original labels. Thus the model sees integer IDs and derived numeric fields, while the raw address and date strings do not enter the encoder.

The flat-model vocabulary is learned from the retained training file and the HDB vocabularies from the block table. Numeric means and population standard deviations are computed over all 149,685 retained training rows before the time split and then applied to both selection and validation rows. These historical preprocessing statistics include the validation period. The strict baseline protocol in Section VI instead fits rental-derived state within the training window.

V. DATA MINING METHODS

The encoded table from Section IV is treated as a short sequence of fields, so the model can learn interactions between location, dwelling attributes and time. The implementation follows the masked-token and self-attention ideas of BERT and the Transformer architecture [1], [2]; its field-wise input construction is also related to FT-Transformer for tabular data [3]. Figure 3 gives an overview: the same BERT backbone is first pretrained to reconstruct masked field tokens and is then fine-tuned with a regression head on `[CLS]` to predict monthly rent.

A. Related Work

Dynamic hedonic models relate housing prices to property attributes while allowing these relationships to vary across time and municipalities [4]. Tabular-learning benchmarks find that tree methods remain competitive on medium-sized datasets, motivating a categorical-feature baseline such as CatBoost [5], [6]. Neural alternatives use different mechanisms: TabM builds parameter-efficient MLP ensembles [7], while TabPFN learns from synthetic tasks and predicts using labeled examples as context [8]. ModernNCA learns representations for neighborhood-based prediction [9]. These three neural alternatives were considered but not run in this study. Our experiments compare the recorded baselines and field-token BERT under future-month protocols, whose rankings must be checked on the rental data itself.

B. Baselines and Design Rationale

The comparison includes four baselines with increasing complexity. The global mean uses the training-period rent average. The recent grouped mean uses the latest six training months and averages by town and flat type, falling back to a flat-type mean and then a global mean when a group has fewer than ten observations. Ridge [10] uses one-hot encodings for the eight categorical IDs with regularization strength 10, while shallow CatBoost [6] uses the same 30 inputs, depth 4, 500 rounds, learning rate 0.1 and seed 42. Its number of rounds is fixed without using validation labels. Section VI evaluates all four baselines under both protocols.

We selected a field-token encoder because self-attention can represent pairwise and higher-order interactions without imposing a linear effect for month, area or distance. Masked-field reconstruction is intended to learn structure from labeled-period features without using rent labels. Its incremental benefit remains untested because BERT was not evaluated without pretraining.

C. Field Tokenization and Encoder

Each row contains 30 scalar inputs: eight categorical IDs, 19 standardized numeric fields, two calendar scalars and one imputation flag. The two calendar values share one token, giving 29 field tokens. Category IDs use learned embeddings (dimensions 16, 16, 32, 8, 8, 8, 8 and 8 in the configured field order) followed by a projection to width 768. Each numeric field uses a separate $1 \to 32 \to 768$ network with GELU; the calendar pair uses a $2 \to 768$ projection and the flag a $1 \to 768$ projection. A learned `[CLS]` vector is prepended, so the encoder sequence has length 30.

The backbone is a BERT encoder with 12 layers, hidden width 768, 12 attention heads, intermediate width 3072 and dropout 0.1. The Hugging Face `BertModel` is instantiated from a configuration, with all encoder weights randomly initialized for this tabular task. Fixed field order and positional embeddings identify which field each token represents. Figure 4 summarizes the token construction, encoder and the two training heads.

D. Masked-Field Pretraining

For every training row, each field group is selected with probability 0.15 and at least one group is forced to be selected. The selected token is replaced by a learned mask vector. Two sets of mutually determined

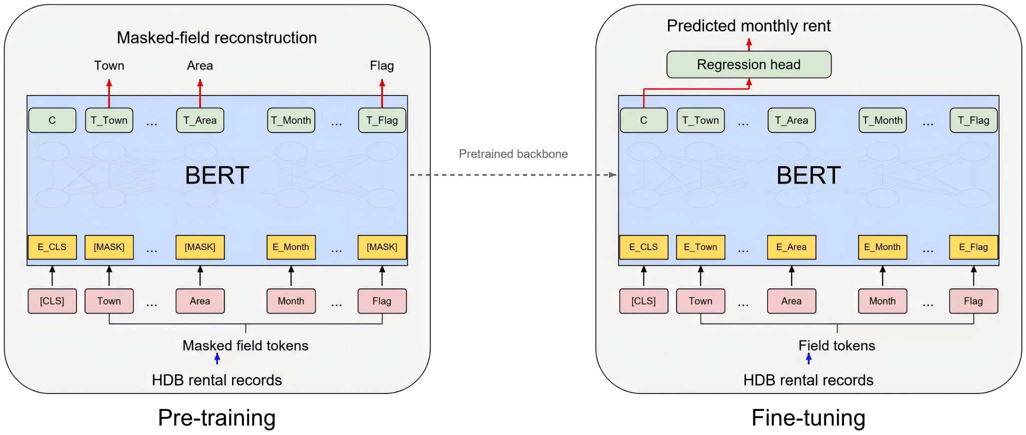
Masked-field reconstruction
Town
Area
Flag
C
T_Town
T_Area
T_Month
T_Flag
BERT
E_CLS
[MASK]
E_Month
[CLS]
Town
Area
Month
Flag
Masked field tokens
HDB rental records
Pre-training
Pretrained backbone
Predicted monthly rent
Regression head
E_Town
E_Area
E_Flag
Field tokens
Fine-tuning

Fig. 3. Overview of the two-stage procedure. Left: during pretraining, randomly selected field tokens of an HDB rental record are replaced by `[MASK]` and the encoder reconstructs them (e.g., town, area, flag); rent is not used. Right: the pretrained backbone is reused, and a regression head on the `[CLS]` output predicts the monthly rent. Figure 4 gives the layer-level details.

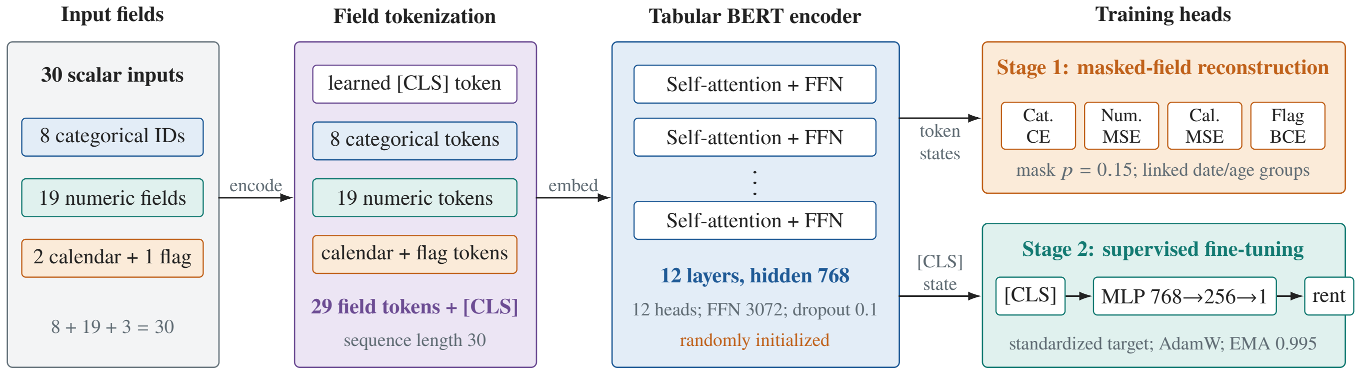
Input fields
30 scalar inputs
8 categorical IDs
19 numeric fields
2 calendar + 1 flag
8 + 19 + 3 = 30
encode
Field tokenization
learned [CLS] token
8 categorical tokens
19 numeric tokens
calendar + flag tokens
29 field tokens + [CLS]
sequence length 30
embed
Tabular BERT encoder
Self-attention + FFN
Self-attention + FFN
Self-attention + FFN
12 layers, hidden 768
12 heads; FFN 3072; dropout 0.1
randomly initialized
token states
[CLS] state
Training heads
Stage 1: masked-field reconstruction
Cat. CE
Num. MSE
Cal. MSE
Flag BCE
mask p = 0.15; linked date/age groups
Stage 2: supervised fine-tuning
[CLS]
MLP 768→256→1
rent
standardized target; AdamW; EMA 0.995

Fig. 4. Two-stage field-token model. Thirty scalar inputs form 29 field tokens and a learned `[CLS]` token. Stage 1 reconstructs masked fields with type-specific heads; Stage 2 reads `[CLS]` with a rent regression head. The encoder is randomly initialized BERT (12 layers, width 768, 12 heads).

fields are masked jointly: `MONTH_INDEX` with the calendar pair, and `LEASE_COMMENCE_DATE`, `AGE_YEARS`, `YEAR_COMPLETED` and `COMPLETION_AGE`. Joint masking prevents the reconstruction task from becoming a direct copy from a visible duplicate field. The category heads use cross-entropy, numeric and calendar heads use mean squared error, and the flag head uses binary cross-entropy; the available type losses are averaged within type and summed. Rent is not passed to this stage.

E. Rent Regression and Training Procedure

For fine-tuning, the `[CLS]` representation is passed through LayerNorm, a 768-to-256 linear layer, GELU, dropout and a final scalar layer. The target is standardized using the current training rows, mean squared error is optimized, and predictions are transformed back to SGD/month before scoring. AdamW uses weight decay 0.01; the pretraining learning rate is 10^{-4}, while fine-tuning uses 2×10^{-5} for the encoder and 10^{-4} for the head. The configured effective batch size is 1,024, gradient norm is clipped at 1.0, and the schedules use two or three warm-up epochs followed by cosine decay. Fine-tuning maintains an exponential moving average with decay 0.995. The recorded BERT runs use seed 42 and float32 arithmetic.

Training is separated into selection and refitting. In the selection round, the first 39 labeled months (2021-01–2024-03) train the model and the last 12 labeled months select the epoch, with a patience of five epochs. The recorded best epochs are 40 for masked-field pretraining and 7 for fine-tuning. The refit round then trains on all 149,685 retained rows for exactly those epoch counts and does not use validation labels. These values describe one frozen configuration with early stopping; they do not constitute a systematic hyperparameter search.

VI. EXPERIMENTAL EVALUATION

The method in Section V is evaluated with a whole-month holdout that approximates the future-oriented test setting. We report historical and strict protocols separately because their cleaning order and validation rows differ.

A. Protocol and Reproducibility

The historical protocol follows the recorded BERT pipeline: the full labeled file is cleaned before the time split. The first 39 months, 2021-01–2024-03, form training; the last 12 months,

TABLE III

Historical-protocol model comparison on 35,058 retained validation rows. Bias is prediction minus the original recorded rent; all metrics are SGD/month.

Model	RMSE	MAE	Bias
Global mean	842.54	686.73	-534.45
Recent 6-month town$\times$flat-type mean	497.97	374.20	-51.09
Ridge	606.82	459.95	$+371.92$
Shallow CatBoost	468.11	348.07	-37.46
Two-stage BERT	468.34	344.83	-6.73

2024-04–2025-03, form validation. The selection run contains 114,627 training rows and 35,058 retained validation rows. The original validation window has 35,158 rows, of which 100 are removed by full-file rent cleaning. The four new baseline runs use this same order and score preserved original rents on the retained rows; the saved BERT summary supplies the fifth result.

The strict protocol splits by month first. It cleans and smooths only training-period rents and fits scaling statistics and the flat-model vocabulary on that period. HDB vocabularies still come from the auxiliary block table. Validation passes through the frozen preprocessing state without labels, retaining all 35,158 rows for scoring against original rents. Only the four baselines have strict results; BERT has not been retrained under this protocol.

RMSE is computed on the SGD/month scale with one equal weight per row; MAE and mean error (prediction minus target) are reported as diagnostics. The recorded BERT run used seed 42, deterministic float32 settings, Python 3.12.14, PyTorch 2.10.0+cu128, Transformers 4.57.6, NumPy 2.3.5, pandas 3.0.1 and an NVIDIA GeForce RTX 5090 Laptop GPU. Its refit uses all 149,685 retained training rows, with 40 pretraining epochs and 7 fine-tuning epochs selected in the preceding round. Code, configurations and recorded summaries are available in the project repository.[1]

B. Model Comparison (RQ1)

Table III compares models under the historical protocol. The recent grouped mean already reduces RMSE from 842.54 to 497.97 SGD/month, indicating that a large part of the predictable variation is captured by recent town–flat-type price levels. Shallow CatBoost and the two-stage BERT are nearly tied in RMSE: their difference is 0.23 SGD/month. BERT has lower MAE and a smaller absolute bias. These are descriptive results from a single historical window, not evidence of a statistically established or general model ranking. The best RMSE improves on the recent grouped mean by only about 30 SGD/month.

The project README records 86.2 million regression-model parameters and about 4 h 45 min for BERT selection plus refitting on a GPU. The historical CatBoost fit-and-predict pass took about 9 s on a CPU, excluding preprocessing. The different hardware and timing scopes preclude a controlled speed comparison.

TABLE IV

Strict-protocol ablation on all 35,158 validation rows (SGD/month). First row: full-input RMSE; remaining rows: ΔRMSE. Deltas are computed before rounding.

Input / removed group	Ridge	CatBoost
Complete input (RMSE)	615.49	475.83
Street and building IDs (ΔRMSE)	$+8.52$	$+2.95$
Ten facility fields (ΔRMSE)	-0.08	$+0.84$
Time group, five fields (ΔRMSE)	$+85.64$	$+232.26$

TABLE V

Historical BERT selection-round scores under two target conventions. Both rows use the 35,058 retained validation rows; RMSE and MAE are SGD/month.

Scoring target	n	RMSE	MAE	Mean error
Cleaned rent	35,058	437.16	317.74	-6.73
Original rent	35,058	468.34	344.83	-6.73

C. Feature-Group Ablation (RQ2)

Table IV reports strict-protocol ablations, each retraining the same model and configuration after removing one group. Removing street and building IDs raises RMSE by 8.52 for Ridge and 2.95 for CatBoost. Removing ten facility fields changes RMSE by less than 1 SGD/month with town, coordinates and other location fields retained. The largest increases follow removal of the time group: +85.64 for Ridge and +232.26 for CatBoost. Because all other groups stay in the model, each Δ measures a group's additional predictive value given the remaining features, not its standalone importance.

The time group contains five fields: month index, two cyclic components, lease-based age and completion age. Its joint effect cannot be assigned to any single field. CatBoost's strict bias shifts from -33.28 to -505.62 SGD/month without this group, approaching the strict global-mean bias of -528.94. This is consistent with losing the ability to track the later rental level, even though time-filtered MRT features can still carry indirect time information. BERT has no feature-group ablation.

D. Target Convention and Within-Input Variation

For completeness, Table V reports the same BERT predictions under the cleaned and preserved original labels. The cleaned target is used for early stopping; the original target is closer to the uncleaned competition label. They are scoring conventions for one historical model, not two methods.

For a group of identical inputs, every model must emit one value; the group mean is optimal for squared error, while deviations of the recorded rents around that mean remain. On this retained validation set, the two recorded RMSE values satisfy the decomposition in Eq. 2:

$$468.3424^2 \approx 437.1584^2 + 168.04^2. \tag{2}$$

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[1] https://github.com/MOYU114/cs-5228-proj. Reproduction commands and environment requirements are in the project and baseline READMEs. Raw course data, checkpoints and BERT row-level predictions are external to Git.

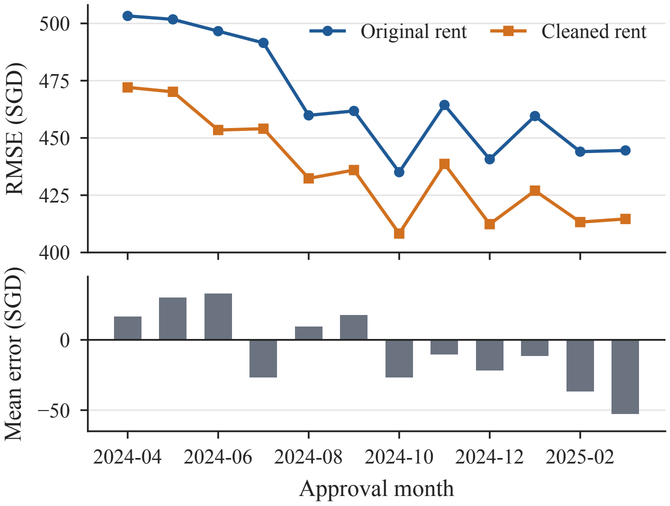

Fig. 5. Historical BERT validation by approval month for the same 35,058 retained rows. The upper panel shows RMSE on cleaned and original recorded rents; the lower panel shows mean error (prediction minus original recorded rent). All values are SGD/month.

Thus 168.04 SGD is the RMS within-input variation isolated by this decomposition. It is a data-dependent diagnostic floor for a model restricted to the supplied fields, not proof that every future row has the same irreducible component.

E. Error Analysis (RQ3)

Figure 5 shows month-wise BERT error. The upper panel contains the two RMSE curves; the lower panel shows monthly mean error as bars. On the original labels, RMSE is high in 2024-04 (503.3 SGD) and 2024-06 (496.6 SGD) and falls below 445 SGD in 2025-02 and 2025-03. Mean error moves from +32.9 SGD in 2024-06 to −52.8 SGD in 2025-03, so the model is initially high and later low relative to the observed labels. This sign change indicates residual temporal drift.

The baseline files provide complementary temporal evidence. In the historical window, shallow CatBoost has lower monthly RMSE in the first five months, whereas BERT is lower in the final seven months. Ridge's historical monthly mean error grows from approximately +218 SGD in 2024-04 to +532 SGD in 2025-03, a pattern consistent with overly strong linear time extrapolation. Figure 6 shows these monthly biases. The recent six-month town–flat-type mean (RMSE 497.97) is also more accurate than Ridge (606.82) in Table III: even with area and location inputs, how the past rent trend is carried into future months strongly affects the estimate.

Building coverage exposes a second generalization boundary. Under the strict protocol, CatBoost RMSE is 475.50 for buildings seen during training and 548.26 for unseen buildings; the unseen group contains only 151 rows. The small unseen sample makes that estimate uncertain. BERT's row-level predictions are unavailable for this building-level breakdown.

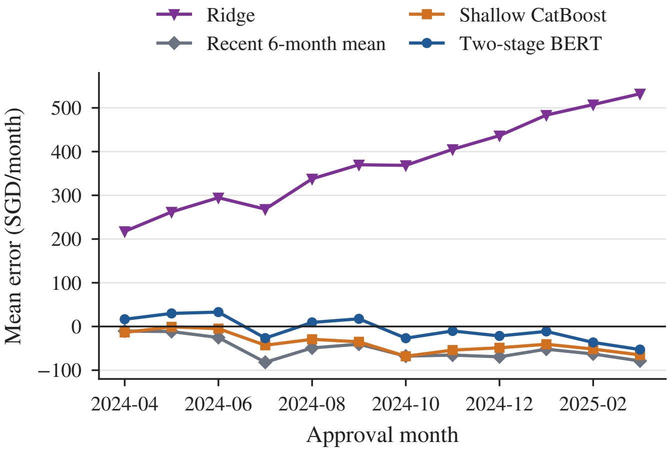

Fig. 6. Monthly mean error (prediction minus original recorded rent) under the historical protocol, 35,058 retained validation rows, SGD/month. Ridge over-predicts increasingly over time, while the other models stay within 85 SGD of zero. The global mean (bias about −500) is omitted for scale.

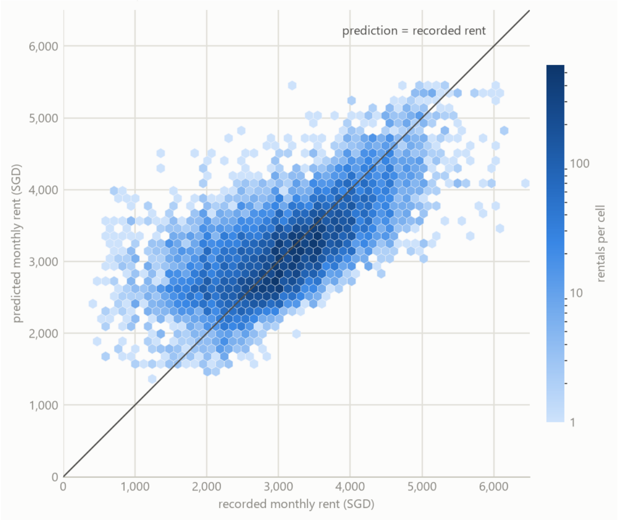

Fig. 7. Predicted versus original recorded validation rent for 35,058 retained rows. Both axes are SGD/month; the diagonal is the identity line.

Figure 7 compares BERT predictions with recorded rents. The diagonal trend is captured, while the prediction range is visibly narrower than the observed range, indicating shrinkage toward typical rents for the represented field combinations. The available fields do not identify the omitted unit attributes behind individual residuals.

VII. DISCUSSION AND LIMITATIONS

A. Answers to the Research Questions

RQ1: Model complexity. The recent town–flat-type mean captures much of the improvement over the global mean; the best historical model improves on the recent-mean RMSE by only about 30 SGD/month. Shallow CatBoost and BERT differ by 0.23 RMSE, with BERT having lower MAE and absolute bias. This single-window result supports a competitive shallow-tree baseline, but does not establish a general model ranking.

RQ2: Feature groups. Time is the most influential tested group for Ridge and CatBoost. The large negative bias after removing it suggests that matching later rent levels matters more than adding facility distances and counts to the existing location inputs. This conclusion is conditional on the retained features and the joint removal of the time group.

RQ3: Error concentration. Ridge increasingly overestimates later months, BERT's bias shifts from positive to negative, and CatBoost has higher RMSE on the small unseen-building sample. BERT predictions also shrink toward typical rents, while within-input variation persists because the recorded fields cannot distinguish every unit. These diagnostics identify temporal calibration, building coverage and missing unit detail as priorities beyond aggregate RMSE.

B. Evidence and Protocol Limitations

BERT has not been evaluated without pretraining or with feature-group removals, leaving the benefit of masked-field reconstruction and its feature sensitivities unresolved. Given the similar historical RMSE and the recorded computational cost, the current evidence provides little RMSE justification for the larger model.

Historical target cleaning precedes the split, and numeric standardization uses all retained labeled rows. Validation-period information can thus affect label-dependent cleaning and feature statistics; the original-label score also excludes the 100 removed validation rows. Strict baselines address this preprocessing order and score all 35,158 original validation rows, but a strict BERT result is still missing.

C. Generalization and Future Work

The test horizon reaches 16 months beyond the training maximum, compared with 12 validation months. Calendar features alone cannot guarantee a correct continuation of the rental level. The test set also contains 102 rows from 50 blocks absent from the full training history. Schools and malls are static snapshots without historical effective dates, and HDB facts omit unit-level floor and condition. The reported associations cannot isolate causal rental drivers.

The next priority is to retrain BERT under the strict protocol, then compare direct fine-tuning without pretraining and the same feature-group removals. Exporting row-level predictions would enable building-level checks. Further temporal validation windows would test whether the observed rankings and calibration patterns persist.

VIII. CONCLUSION

We studied future-month HDB rent prediction through EDA-guided preprocessing, baseline comparisons and a two-stage tabular BERT. In the historical window, BERT achieves 468.34 SGD/month RMSE and CatBoost 468.11, with BERT having lower MAE and absolute bias. The recent grouped mean is already within about 30 SGD/month RMSE of the best model. Strict baseline ablations identify the time group as the most influential tested group, while monthly bias, unseen-building errors and the 168.04 SGD within-input variation component explain limitations hidden by the overall score. Strict BERT evaluation and a no-pretraining comparison remain the priorities for testing the value of the two-stage approach.

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APPENDIX A
TEAM AND CONTRIBUTION STATEMENT

GROUP 58 members are listed below in ascending student-ID order. The completed division of work will be added after the team confirms it.

TABLE VI
TEAM MEMBERS AND CONTRIBUTION STATUS

Member	Student ID	Contribution
CHEN CONGRUI	A0329163A	Pending confirmation
ZHU JIFENG	A0351860E	Pending confirmation
JIN BINGHAN	A0353576U	Pending confirmation
SHAO GUJIE	A0356273Y	Pending confirmation